6.Implementation of Hierarchical Clustering

Aim:

To write a program that implements hierarchical (agglomerative) clustering to group the given data points into clusters based on their similarity, and to display the cluster labels and a dendrogram showing how the clusters are merged.

Algorithm :

1. Start the program.
2. Import the required libraries (pandas, matplotlib, dendrogram and linkage from scipy.cluster.hierarchy, and AgglomerativeClustering from sklearn.cluster).
3. Load the dataset from the input file.
4. Select the input features to be used for clustering (no target variable is needed, since clustering is unsupervised).
5. Treat each data point as its own separate cluster.
6. Compute the distance between every pair of clusters (for example Euclidean distance with
Ward linkage).
7. Merge the two closest clusters into a single cluster.
8. Update the distances between the new cluster and all the remaining clusters.
9. Repeat steps 7 and 8 until all the data points are merged into one cluster.
10. Plot the dendrogram to show the merging process and decide the number of clusters by
cutting the tree at a suitable height.
11. Create the AgglomerativeClustering model with the chosen number of clusters (for example
3) and fit it on the data to obtain the cluster label for each data point.
12. Display the cluster labels and add them to the dataset.
13. Plot the data points coloured by cluster (optional).
14. Stop the program.

Program,Result&Graph:

```
import pandas as pd
import matplotlib.pyplot as plt
from scipy.cluster.hierarchy import dendrogram, linkage
from sklearn.cluster import AgglomerativeClustering

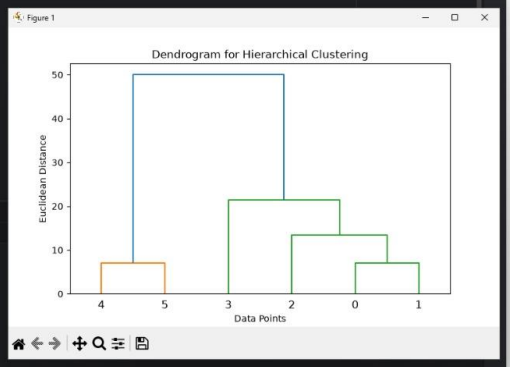
# Load dataset
data = pd.read_csv("data.csv")

# Select features
X = data[['Hours_Studied', 'Marks']]

# Create linkage matrix
linked = linkage(X, method='ward')

# Plot Dendrogram
plt.figure(figsize=(8, 5))
dendrogram(linked)
plt.title("Dendrogram for Hierarchical Clustering")
plt.xlabel("Data Points")
plt.ylabel("Euclidean Distance")
plt.show()

# Apply Hierarchical Clustering
hc = AgglomerativeClustering(n_clusters=3)
```




```
C:\Users\Admin\PycharmProjects\PythonProject18\venv\Scripts\python.exe C:\Users\Admin\PycharmProjects\PythonProject18\150.py
```

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import pandas as pd
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from scipy.cluster.hierarchy import dendrogram, linkage
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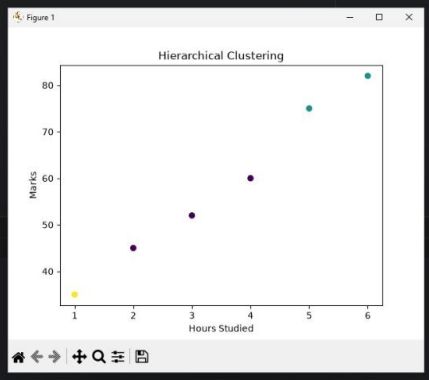
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```
C:\Users\Admin\PycharmProjects\PythonProject18\venv\Scripts\python.exe C:\Users\Admin\PycharmProjects\PythonProject18\150.py
   StudentID  Hours_Studied  Attendance  Assignments  Marks Result  Cluster
0          1              2          75            3     45   Fail        0
1          2              3          80            4     52   Pass        0
2          3              4          85            5     60   Pass        0
3          4              1          65            2     35   Fail        2
4          5              5          90            5     75   Pass        1
5          6              6          95            5     82   Pass        1
```